

Inbox Avenger Final Status Report

Jonathan Cagle, Benjamin Marshall, Jordan Wishom

Why We Chose The Inbox Avenger

We chose the Inbox Avenger because many people in our lives and around the world have way too much spam and scam emails in their Inbox.

Many users do not have the time or patience to work and manage an email inbox, this tool was designed to be easy to use in order to manage this problem, for users with any range of computer skill level.

We wanted to develop a tool that mitigated this, and create something that every single email user could use in their daily lives, that would allow users to take back their inbox entirely.

Version 1 Initial Implementation (Inbox Avenger Lite)

- Known as the Inbox Avenger Lite
- Scrapped multiple rows and badged certain emails as:
 - High Risk
 - Promo
 - Low Priority
- Very basic structure and prone to badging errors
- Very small word Database to identify patterns
- Limited on screen search, with no full email search

Version 2 Adaptive Version

- Transitioned from basic scanning to a more interactive system
- Introduced dynamic user interface:
 - Floating toolbar
 - Settings panel within Gmail
- Added customizable features:
 - Accent colors & category colors
 - Font and theme controls
 - Scan depth and sorting options
- Improved classification logic:
 - Expanded keyword database
 - Regex-based pattern detection
 - Weighted scoring system
- Implemented adaptive learning system:
 - User-driven “teach” functions
 - Local pattern storage for future scoring
- Shifted toward a more scalable and user-focused design

Last Known Limitations from Version 3

- It does not have full Gmail API access
 - This means it can only scan:
 - What is seen on screen
 - What is visible to the user
 - What is opened on screen
 - What Unsubscribe links are visible to the current message
- It cannot yet:
 - Scan the full inbox history directly
 - Apply true Gmail labels across multiple accounts
 - Guarantee unsubscribing handling for every sender automatically

Previous Next Steps and Aim of Final Direction

- The next steps are:
 - Continue using the Chrome Extension model as it works quite well
 - Add more automation for emails that are not just visible just on the screen
 - Improve the scoring accuracy and whole message analysis
 - Expand the subcring management features
 - Build toward smarter machine-learned auto-unsubscribe features
 - Later intend to implement machine learning to improve classification and automate unsubscribe decisions more intelligently than now
 - Eventually move toward Gmail API integration for deeper mailbox wide control
 - Explore security measures as the system advances
 - Explore possible need for database storage for saved email patterns, user preferences, and future scalability
- We will continue the use of Chrome Extension because it is the best consistent way to keep building the concept, while giving users the ability to have interactive experience

Where We Left Off After Version 3

- At the end of Version 3, the prototype had moved beyond basic row scanning
- Version 3 could inspect opened email threads and review more complete email content
- The system was able to look at:
 - Sender information
 - Subject information
 - Snippets
 - Full body text
 - Links inside opened emails
- This helped improve detection for risk, newsletters, promotional emails, and useful deals
- However, the prototype was still limited by what Gmail made visible on screen

Version 3 Next Steps Going Forward

- After Version 3, the main next step was to make the extension smarter and more organized
- The prototype needed stronger scoring logic instead of relying mostly on visible text and simple patterns
- Important next goals included:
 - Improving email classification accuracy
 - Expanding the word database
 - Making categories more specific
 - Improving subscription and unsubscribe management
 - Adding more user control over how emails are labeled
 - Preparing the project for deeper automation in the future
- The goal was to move from a review tool into a more complete inbox assistant

Version 3 to Version 4: Main Direction Change

- Version 4 focused on improving the intelligence behind the prototype
- Version 3 could scan more email content, but Version 4 focused more on how that content was interpreted
- The biggest change was improving how the extension categorized emails after scanning them
- Version 4 expanded the system from simple classification into a more personalized scoring tool
- This helped the prototype feel more adaptable instead of only relying on fixed rules

Version 4 Code Changes: Expanded Word Database

- Version 4 added a stronger and more organized word database
- The database was expanded to support more accurate classification
- The code separated words and phrases into clearer groups such as:
 - High-risk terms
 - Scam terms
 - Spam terms
 - Promotional terms
 - Useful deal terms
 - Safe business or transactional terms
 - Unsubscribe-related terms
- This made the scoring system more explainable because each email could be labeled based on specific matched signals

Version 4 Code Changes: Sender Trust and Blocking

- Version 4 added stronger sender-based logic.
- The prototype could now consider whether a sender was trusted, unknown, or blocked.
- Trusted senders could lower the risk score if the email looked normal.
- Blocked or suspicious senders could increase the risk score.
- This was important because the sender is a major factor in deciding whether an email is safe.
- This made the extension act more like a real inbox security and organization tool.

Version 4 Code Changes: Adaptive Learning Improvements

- Version 4 improved the adaptive learning concept from earlier versions.
- The system could save user corrections and learned patterns locally.
- Learned words could be stored into custom categories such as custom good words, custom bad words, and custom deal words.
- This allowed the prototype to become more personalized over time.
- Instead of only using the original word database, the extension could adjust based on how the user corrected emails.

Version 4 to Version 5: Final Prototype Upgrade

- Version 5 was created to make Inbox Avenger more reliable, complete, and able to be demonstrated
- Version 4 had strong features, but Version 5 focused on cleaning up the system and making it feel like a final prototype.
- The main goals for Version 5 were:
 - Clean up the code structure
 - Improve stability inside Gmail
 - Keep the major features from previous versions
 - Add a small amount of machine learning
 - Improve the dashboard and scan history
 - Make the project easier to present during the final demonstration

Version 5 Code Changes: Rule System + Small Machine Learning

- Version 5 added a lightweight machine-learning feature to support classification
- The system uses a small local Naive Bayes style classifier
- This does not replace the rule-based system but works with it
- Final classification is based on both:
 - Keyword and phrase matches
 - Risk scoring
 - Sender trust/blocking
 - Link detection
 - Machine-learning confidence
- This made the prototype more advanced while still keeping it simple enough to run locally

Version 5 Code Changes: Automation and Demonstration Features

- Version 5 added more automation and presentation-ready tools.
- The extension now includes Demo Mode, a manual email analyzer, a visible Gmail row scanner, background scheduled scan structure, scan history, dashboard statistics, correction learning, and import/export settings.
- Demo Mode is important because it allows the project to work immediately without needing Google API setup.
- This makes Version 5 easier to demonstrate and more reliable in demonstrative cases

Version 5 Code Changes: Gmail Stability Improvements

- Version 5 also fixed issues from earlier Gmail scanning versions.
- Earlier versions had problems where risk panels or badges could reload too often.
- Version 5 made Gmail scanning more conservative and stable.
- The extension now avoids repeatedly triggering itself when badges are added.
- Visible Gmail scanning is used as a fallback method.
- The inbox avenger also includes a Gmail API path for future full inbox scanning.
- This helps separate what works now from what can be expanded later.

Version 5 to Version 6: Future Potential Capabilities

- Version 6 would move Inbox Avenger closer to a production-level tool.
- The biggest future upgrade would be Google OAuth login and Gmail API access.
- With Gmail API access:
 - The extension could scan emails beyond what is visible on screen with ease
 - Scan multiple pages of the inbox automatically
 - Apply real Gmail labels to messages
 - Improve background scanning,
 - Support deeper inbox cleanup and reporting
- This would make the tool more powerful, but it would also require more privacy and security planning.

Version 6 Future Features and Final Direction

- Conclusively Version 6 could expand the project with stronger automation and intelligence.
- The future features include a stronger trained machine-learning model, safer unsubscribe preview and bulk cleanup tools, full inbox scanning through Gmail API, secure storage for user preferences and learned patterns, more advanced dashboard reports, a better user feedback training system, and stronger privacy controls for email data.
- Overall we can see that Version 5 proves the core idea works as a Chrome extension.
- If this was continued Version 6 would focus on making the system more complete, secure, and production-ready.

Thank You!